Delta Physics

Select a topic of your Choice

5 minutes – Individual

Gravitation

Why does the Moon take a month to go round?



Higher Physics

PART 3

Beyond Higher and Exam Practice

The Space Station goes round the Earth in 90 minutes. Why does the Moon take a month?

By the end of this part…



1. I can **use g = GM/r²** (beyond Higher)
2. I can **describe** how **g** changes with **height**
3. I can **explain** why **higher orbits** are **slower**
4. I can **answer** exam questions on **$F = Gm_1m_2/r^2$**

Where g Comes From

Beyond Higher — not in the exam

Weight

$W = mg$

Newton's law

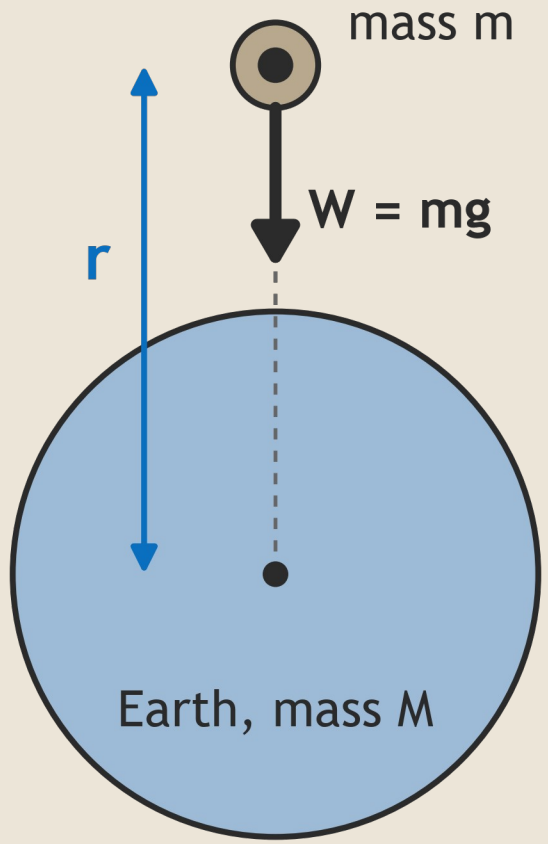
$W = GMm/r^2$

So

$mg = GMm/r^2$: **m cancels**

Result

$g = GM/r^2$



r: from the centre

Field Strength g

Beyond Higher — not in the exam

$$g = (G\,M)/r^2$$

g

field strength (N kg^{-1})

G

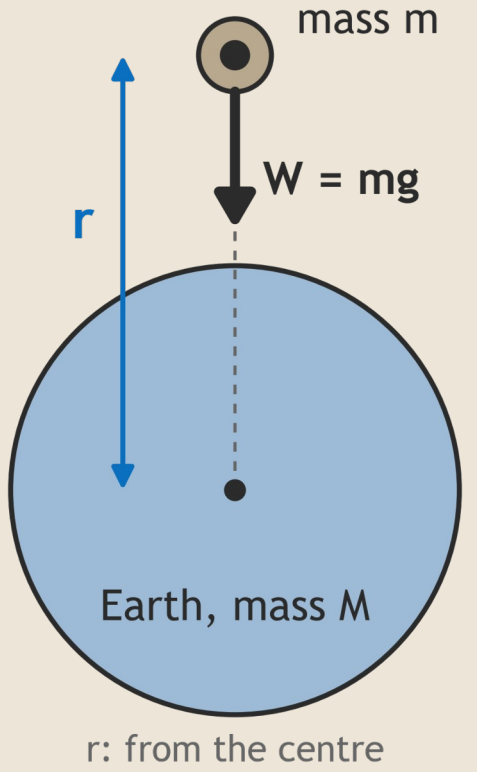
on the data sheet

M

planet's mass (kg)

r

from the centre (m)



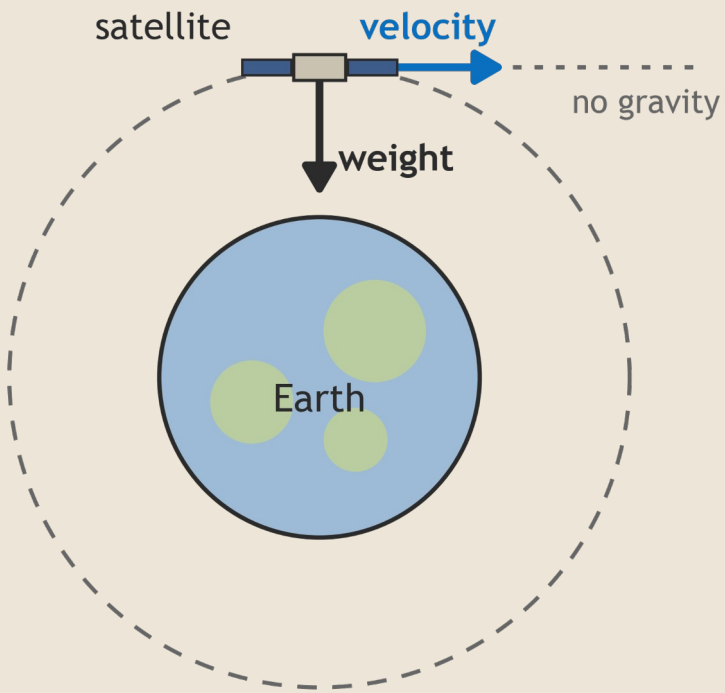
Example: g at the ISS

Beyond Higher — not in the exam

The ISS orbits 6.82×10^6 m from the centre of the Earth.

The mass of the Earth is 6.0×10^{24} kg.

Calculate the gravitational field strength at the ISS. **3**



Example: Marking

Beyond Higher — not in the exam



Question	Expected response	Max mark	Additional guidance
	$g = GM/r^2$ (1 mark) $g = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} / (6.82 \times 10^{6})^2$ (1 mark) $g = 8.6$ N kg^{-1} (1 mark)	3	About 90 % of 9.8 N kg^{-1}: gravity has not gone (Part 1)

g Against Distance

Beyond Higher — not in the exam

Surface

9.8 N kg^{-1}

ISS

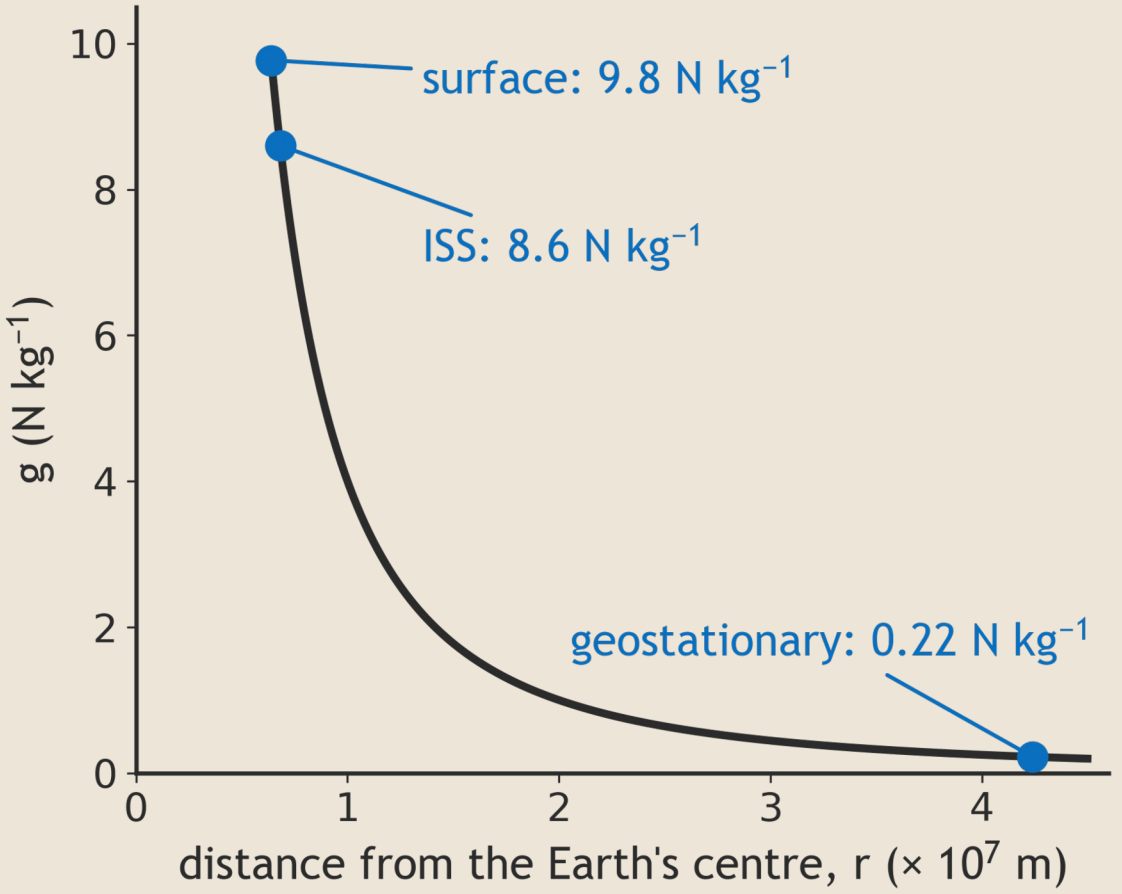
8.6 N kg^{-1}

Geostationary

0.22 N kg^{-1}

Double r

g ÷ 4: inverse square



Orbital Speed

Beyond Higher — not in the exam

$$v = \sqrt{((G\,M)/r)}$$

v
orbital speed (m s^{-1})

G
on the data sheet

M
planet's mass (kg)

r
orbit radius (m)



Orbital Period

Beyond Higher — not in the exam

	ISS	Geostationary
r (m)	6.82×10^6	4.24×10^7
v ($m\ s^{-1}$)		
T		

Speed

$v = \sqrt{(GM/r)}$

One lap

$T = 2\pi r/v$

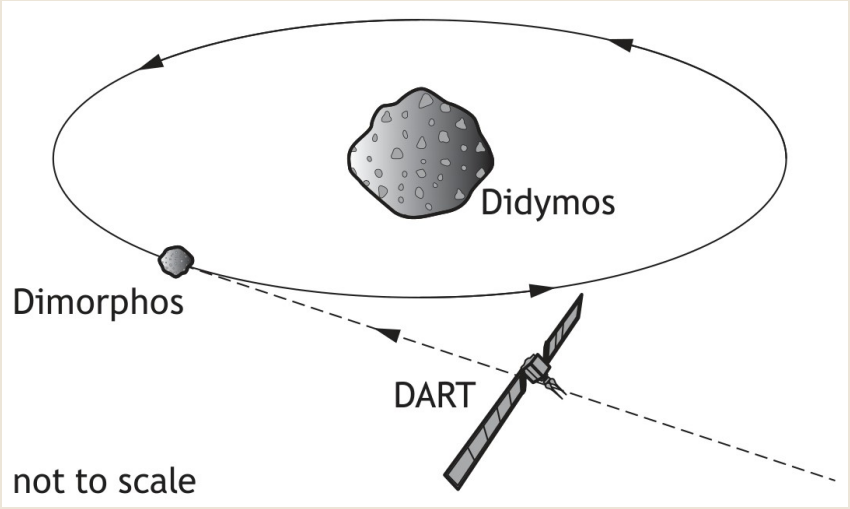
24 h

keeps pace with the Earth: **dishes do not move**

Past Paper: 2024 Paper 2 Q4(a)(ii)

4. (a) The mass of Didymos is taken to be 5.3×10^{11} kg.
The mass of the Earth is 6.0×10^{24} kg.

(ii) When DART collided with Dimorphos, the distance between Earth and Didymos was 1.1×10^{10} m.
Determine the gravitational force between Earth and Didymos.



3

Scan: page 15.

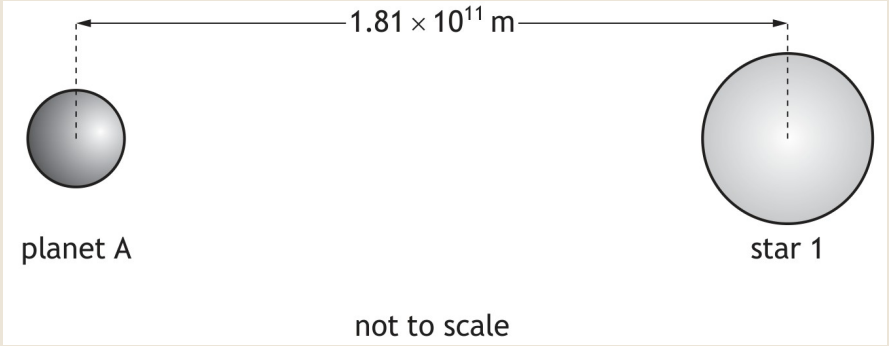
Marking: 2024 Paper 2 Q4(a)(ii)

Question	Expected response	Max mark	Additional guidance
4 (a)(ii)	$F = Gm_1m_2/r^2$ (1 mark) $F = 6.67 \times 10^{-11} \times 5.3 \times 10^{11} \times 6.0 \times 10^{24} / (1.1 \times 10^{10})^2$ (1 mark) $F = 1.8 \times 10^{6}$ N (1 mark)	3	Accept: 2, 1.75, 1.753

Past Paper: 2025 Paper 2 Q5(c)(i)

5. (c)

(i) One exoplanet, planet A, has a mean orbital radius of 1.81×10^{11} m and a mass of 6.09×10^{24} kg.
The mass of star 1 is 1.77×10^{29} kg.
Calculate the gravitational force between planet A and star 1.



3

Scan: page 18.

Marking: 2025 Paper 2 Q5(c)(i)

Question	Expected response	Max mark	Additional guidance
5 (c)(i)	$F = Gm_1m_2/r^2$ (1 mark) $F = 6.67 \times 10^{-11} \times 6.09 \times 10^{24} \times 1.77 \times 10^{29} / (1.81 \times 10^{11})^2$ (1 mark) $F = 2.19 \times 10^{21}$ N (1 mark)	3	Accept: 2.2, 2.195, 2.1946

Past Paper: 2025 Paper 2 Q5(c)(ii)

5. (c) (continued)

(ii) A second exoplanet, planet B, has three times the mass of planet A.
The mean orbital radius of planet B is double the mean orbital radius of planet A.
State how the gravitational force of attraction between planet B and star 1 compares to the gravitational force of attraction between planet A and star 1.
Justify your answer.

3

Marking: 2025 Paper 2 Q5(c)(ii)

Question	Expected response	Max mark	Additional guidance
5 (c)(ii)	Force is less/¾ (1 mark) Three times the mass gives three times the force. (1 mark) Doubling the distance gives quarter of the force. (1 mark)	3	JUSTIFY Look for this statement first - if incorrect or missing then 0 marks. Do not accept: decrease by ¾ Can justify by calculation: 1.65×10^{21} N Accept: 1.6, 1.646, 1.6460

What Candidates Get Wrong

Substituting

square r; use G, not g (2024 Q4(a)(ii))

Statement first

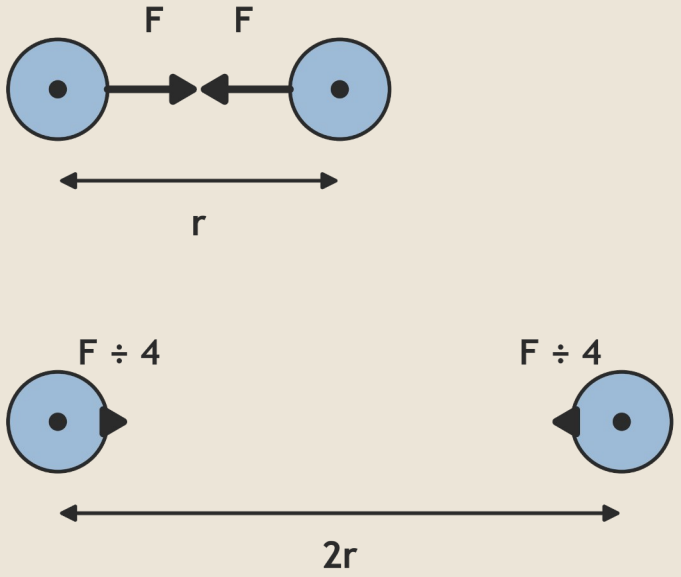
or 0 marks (2025 Q5(c)(ii))

Justify

only some did (2025 Q5(c)(ii), 46 %)

Wording

'decrease by ¾': 0 (2025 MI)



Look back — by the end of this part…



1. I can **use g = GM/r²** (beyond Higher)
2. I can **describe** how **g** changes with **height**
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4. I can **answer** exam questions on **$F = Gm_1m_2/r^2$**

Course specification — 1 of 1

- Knowledge that satellites are in free fall around a planet/star.
- Use of Newton's Law of Universal Gravitation to solve problems involving force, masses and their separation.

Scan: the course specification, page 8.

Credits

Title: the ISS: NASA

Past Paper: 2024 Paper 2 Q4(a)(ii): © Qualifications Scotland (SQA). 2024 Higher Physics Paper 2, Q4

Marking: 2024 Paper 2 Q4(a)(ii): © Qualifications Scotland (SQA). 2024 Higher Physics Paper 2, Q4

Past Paper: 2025 Paper 2 Q5(c)(i): © Qualifications Scotland (SQA). 2025 Higher Physics Paper 2, Q5

Marking: 2025 Paper 2 Q5(c)(i): © Qualifications Scotland (SQA). 2025 Higher Physics Paper 2, Q5

Past Paper: 2025 Paper 2 Q5(c)(ii): © Qualifications Scotland (SQA). 2025 Higher Physics Paper 2, Q5

Marking: 2025 Paper 2 Q5(c)(ii): © Qualifications Scotland (SQA). 2025 Higher Physics Paper 2, Q5

Course specification — 1 of 1: © Qualifications Scotland (SQA). Higher Physics Course Specification v4.1 (September 2026), p.8

R Stewart - November 2026